

GeoAdaptNet: a compact geometric decoder for motor-imagery EEG

Architecture, evaluation, and an honest accounting of what works

Summary

GeoAdaptNet is a 9,678-parameter decoder for binary left/right-hand motor imagery. It classifies filter-bank spatial covariance matrices on the SPD manifold and is designed for **online, label-free deployment**: weights are frozen at test time while two bounded causal states — a covariance recenter and an independent scalar decision-boundary recenter — track session drift without ever seeing a label. A separate intent/rest head lets it abstain rather than emit a command it is unsure of.

Under a leakage-controlled chronological protocol it reaches **86.6%** balanced accuracy (**85.0%** subject-held-out), competitive with strong classical baselines and the best of six leading deep architectures on this cohort, at a fraction of their parameters and 4.5 ms inference. Its distinctive measured property is not peak accuracy but **operating-point safety**: it commits on rest windows 3x less often than any baseline. This report documents the architecture, the evaluation, every lever tested (including the ones that failed), and the limits of what the results support.

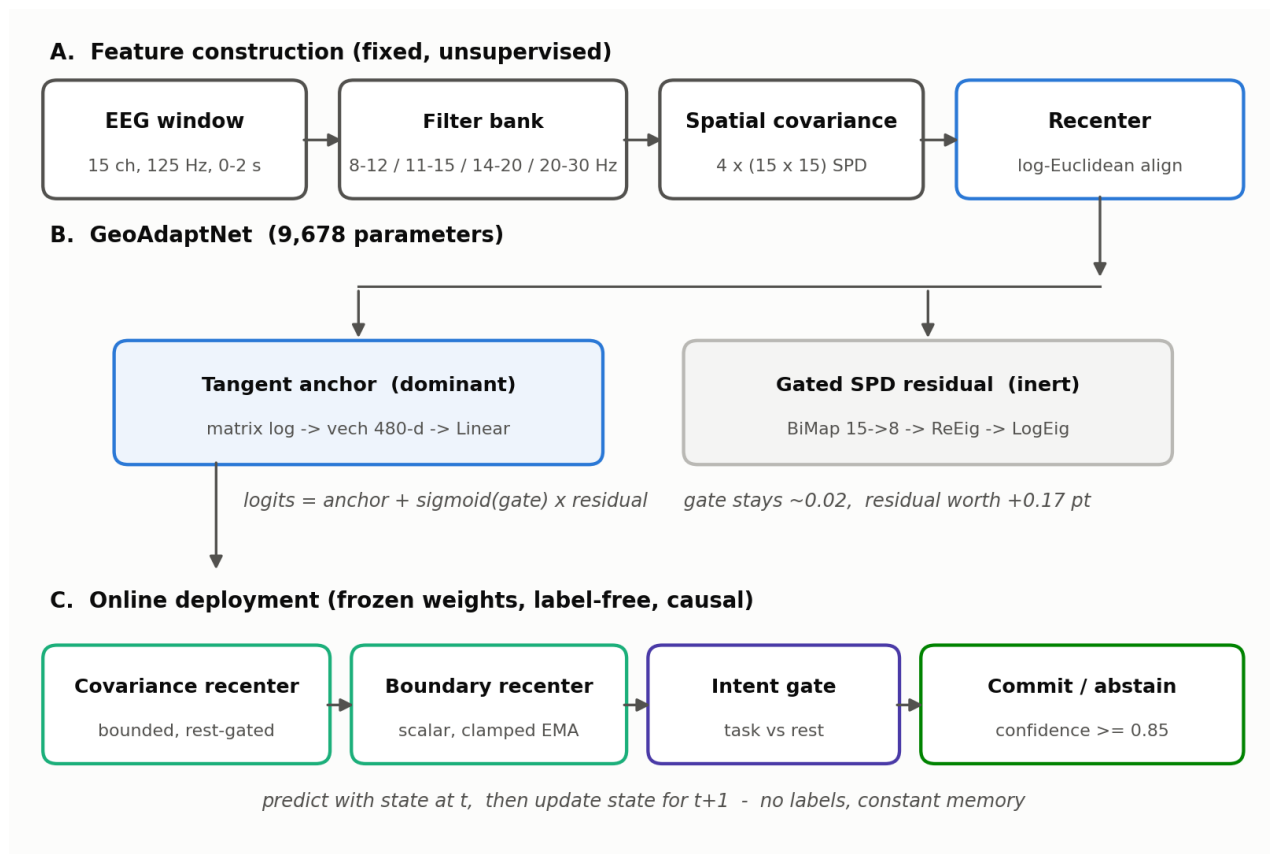


Figure 1 — End-to-end pipeline. Feature construction is fixed and unsupervised; the network is an anchor-plus-gated-residual over SPD matrices; deployment is frozen-weight and causal.

1. Data and feature construction

The cohort is 8 usable participants (IDs 1, 3, 4, 5, 6, 7, 8, 10) x 4 recordings = 32 sessions, acquired on a 15-channel 10-20 montage at 125 Hz. Each cue yields a 0–2 s window (251 samples) that is band-pass filtered into four overlapping bands and reduced to one spatial covariance matrix per band. The classifier therefore never sees raw time series — its input is four 15x15 symmetric positive-definite matrices, a representation that is compact, rotation-aware, and highly data-efficient.

Property	Value
Participants / recordings	8 / 32 (subject 9 and S10 runs 1-4 excluded)
Sampling rate, channels	125 Hz, 15 (10-20 sensorimotor montage)
Window	0.0-2.0 s from cue, 251 samples
Filter bank	8-12, 11-15, 14-20, 20-30 Hz
Model input	4 x (15 x 15) SPD covariance, shrinkage 1e-3
Labelled windows	~1,900 total (~120 per training fold)

Table 1 — Fixed data contract. Every experiment in this report uses it unchanged.

2. Architecture

Anchor (the dominant path). Each band covariance is recentered to a reference by a log-Euclidean congruence, mapped to the tangent space by the matrix logarithm, and vectorized (upper triangle, norm-preserving) into a 480-dimensional feature. A single linear layer produces two logits. This is a geometrically principled, strongly-regularized classifier and it carries essentially all of the model's discriminative power.

Gated SPD residual. In parallel, each band passes through a semi-orthogonal BiMap projection (15 → 8), eigenvalue rectification (ReEig), a matrix log (LogEig), and attention-weighted fusion, producing a correction that is added through a sigmoid gate initialized near zero: $\text{logits} = \text{anchor} + \text{sigmoid}(\text{gate}) \times \text{residual}$. The near-zero gate is a deliberate safety property — the network provably degrades to the classical tangent classifier rather than to an untrained deep path.

Measured finding: the residual is inert. The gate never leaves its initialization (~0.02) and the residual is worth +0.17 points. This is not a training bug. We revived it three ways — auxiliary deep supervision, raising and freeing the gate (it does open, to 0.21), and warm-starting each BiMap with supervised CSP filters — and accuracy did not move. The rank-8 projection is a compressed view of the same covariance the full-rank anchor already reads, so it is structurally redundant.

Intent head. A separate scalar head estimates task-vs-rest. It never turns the two-class problem into a three-class one; it exists to gate commands.

3. Online deployment

At deployment the weights are frozen. An unlabeled chronological prefix (20 windows) seeds two bounded, constant-memory causal states, and every window is *predicted before* it may update them:

State	What it fixes	Update rule
Covariance recenter	input distribution shift	robust log-Euclidean, rest-gated, bounded
Boundary recenter	classifier operating-point drift	slow clamped EMA on rest-like windows
Intent gate	commands during rest	task/rest probability threshold
Commit rule	low-confidence commands	confidence ≥ 0.85 , else abstain

Table 2 — The deployed adaptation stack. No labels, no gradient updates, constant memory.

Decoupling the two states is the design's central claim: aligning covariances does not by itself fix a classifier's threshold, which was observed to drift and skew across sessions. Measured batch-one latency is 4.54 ms (range 4.36-5.71) on an RTX 5070 after covariance construction.

4. Evaluation protocol

Two leakage-controlled protocols are used. **Chronological cross-session:** recordings 1-2 fit the model, recording 3 selects the checkpoint and calibration, and the untouched recording 4 is scored after an unlabeled prefix that is excluded from scoring; neural results average seeds 7/17/27 within participant before cohort statistics. **Subject-held-out (LOSO):** all four recordings of the outer participant are withheld, one further participant is the protected inner validation, and the remaining six fit the model. No target label is ever used for fitting, selection, or adaptation, and every aggregate is recomputed from retained per-window traces.

5. Headline results

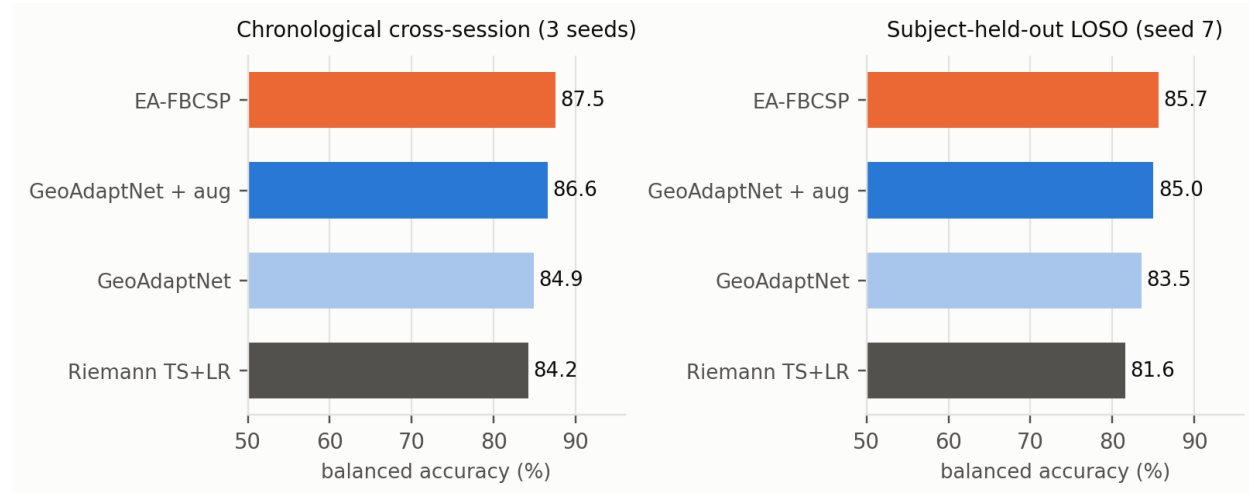


Figure 2 — Balanced accuracy under both protocols. '+ aug' is the improvement adopted in this work.

Model	Chronological	LOSO	Coverage	Rest false-commit
EA-FBCSP (classical)	87.53	85.67	69.4	50.9
GeoAdaptNet + augmentation	86.62	85.02	42.3	19.8
GeoAdaptNet (baseline)	84.86	83.54	41.0	15.7
Riemann TS + LR (classical)	84.24	81.61	67.2	48.0

Table 3 — Balanced accuracy (%), participant-aggregated. Coverage and rest false-commit are from the chronological protocol at a fixed 0.85 commit threshold.

The honest reading: on task accuracy GeoAdaptNet is **competitive, not superior** — it edges the Riemannian baseline and trails EA-FBCSP by under a point. Where it separates is the operating point.

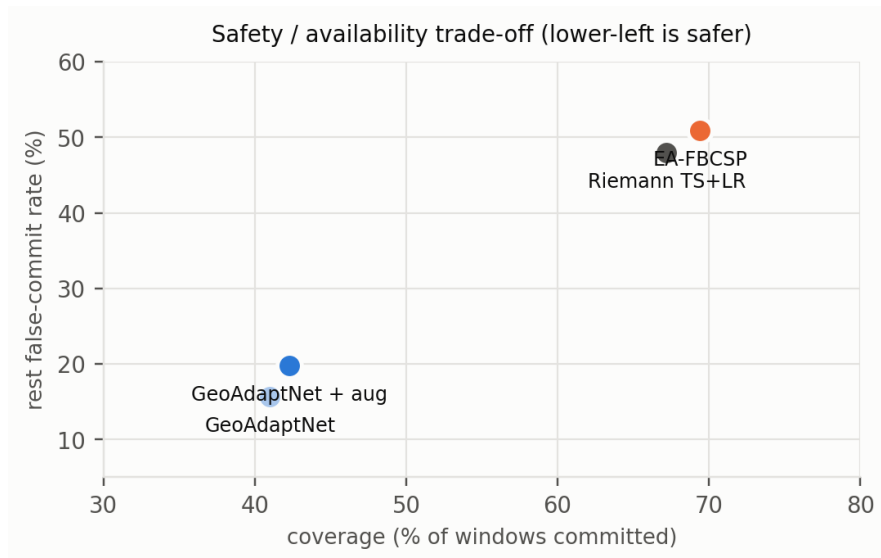


Figure 3 — GeoAdaptNet commits far less often on rest windows, at the cost of lower coverage.

This is a complete-decoder comparison, not proof that the left/right classifier is inherently safer: the baselines have the same confidence threshold but no separate intent head. The safety advantage is bought with availability, and that trade must be stated whenever it is claimed.

6. The improvement adopted

Because the residual is inert, the model is effectively a 480-dimensional linear classifier fitted on roughly 120 training windows per fold — an overfitting regime. The lever that works is therefore regularization, not more depth. **Left/right electrode-swap augmentation** mirrors the sensor montage ($C3 \leftrightarrow C4$, $F3 \leftrightarrow F4$, ...) and flips the label, exploiting the lateral organization of motor imagery; it is a symmetric permutation of the covariance, so it stays exactly on the SPD manifold. Paired with an accuracy-targeted checkpoint-selection metric it gives:

Protocol	Baseline	Improved	Delta
Chronological (3 seeds)	84.86	86.62	+1.76
Subject-held-out LOSO	83.54	85.02	+1.48

Table 4 — The adopted improvement. Selected on the validation recording, not on test; 7 of 8 LOSO participants improve.

The augmentation is architecture-agnostic and helps every network tested (EEGNet gains 13 points from it), so it is a property of the problem's symmetry rather than of this model.

7. Comparison with leading EEG architectures

GeoAdaptNet was benchmarked against five widely-cited braindecode architectures under one uniform protocol — same splits, same metric, every network given the same augmentation, no online adaptation for anyone — so the comparison isolates the architecture. It was then repeated on an entirely independent cohort, Cho2017 (GigaDB 100295) — all 52 subjects, recorded on a different 64-channel amplifier at 512 Hz and mapped onto the same 15-channel / 125 Hz layout, to test whether the local ranking is an artifact of the local recordings.

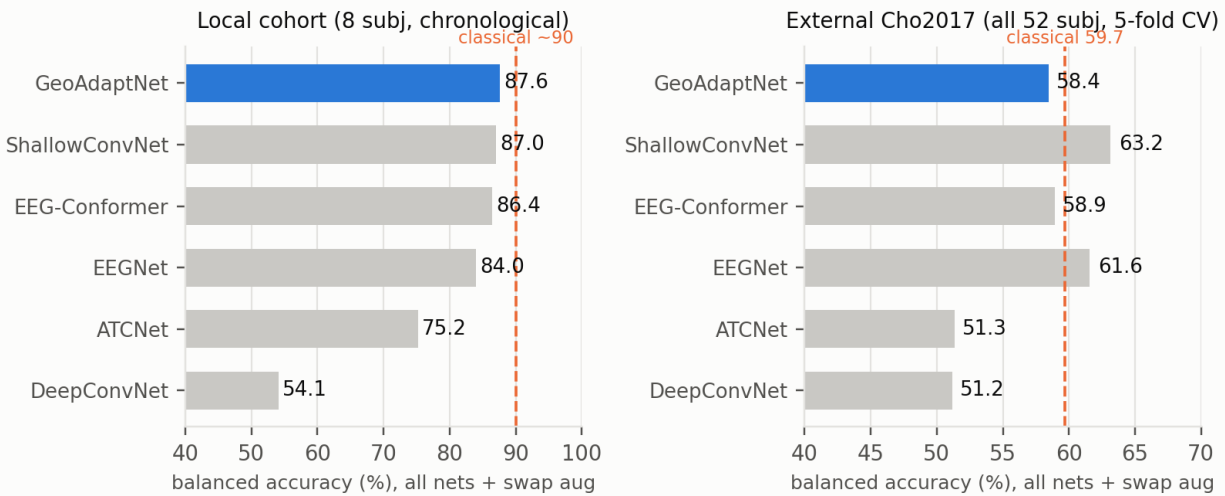


Figure 4 — Architecture comparison, local cohort and the full 52-subject external cohort.

Architecture	Params	Local	Cho2017
GeoAdaptNet	9,645	87.58	58.44
ShallowConvNet	26,002	87.04	63.16
EEG-Conformer	266,306	86.40	58.91
EEGNet	1,602	83.96	61.58

ATCNet	28,868	75.25	51.33
DeepConvNet	233,652	54.10	51.16
Convex-head anchor (this work)	481	88.46	59.59
EA-FBCSP / Riemann (classical)	n/a	90.0	59.7

Table 5 — Balanced accuracy (%), networks with augmentation. Cho2017 is within-subject 5-fold CV over all 52 subjects (none skipped); absolute values are lower because that cohort is far noisier.

On the local cohort GeoAdaptNet is the most accurate deep architecture at the smallest parameter count, and the large data-hungry networks collapse toward chance on ~ 120 training windows — the data-efficiency argument for geometric decoders. **Externally the ranking does not fully hold:** ShallowConvNet leads by 4.7 points over all 52 subjects. GeoAdaptNet remains in the strong group and has the lowest cross-subject variance of any model (8.5 vs ShallowConvNet 11.5), so the local result is not inflated — but it is not the best decoder at that data scale. Notably the 481-parameter convex-head anchor reaches 59.59, statistically indistinguishable from the classical Riemannian pipeline (59.68): that is the fixed-band geometric ceiling, and it sits ~ 3.3 points below ShallowConvNet.

8. Every lever tested

The following were implemented and measured under the same protocol. Reporting the failures is the point: they bound what the architecture can be claimed to do.

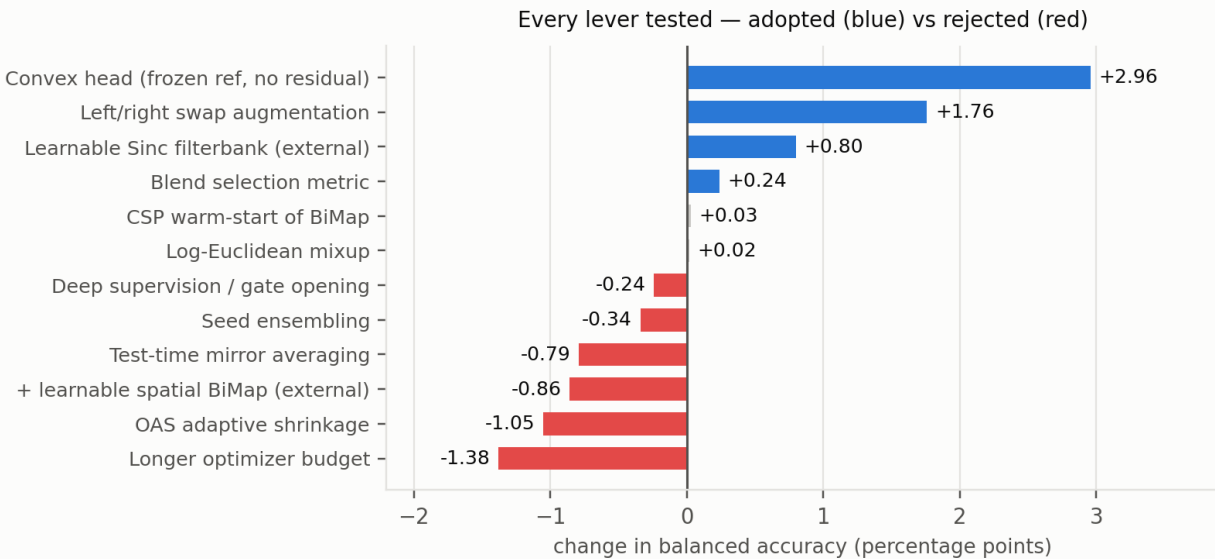


Figure 5 — Change in balanced accuracy for every lever tested. Only two are adopted.

Lever	Result	Verdict
Left/right swap augmentation	+1.76 pt	adopted
Convex head, frozen reference, no residual	+2.96 pt (local)	adopted
Deep supervision / opening the residual gate	no change	rejected
CSP warm-start of the BiMap filters	no change	rejected
Log-Euclidean covariance mixup	no change	rejected
Seed ensembling (3 seeds)	-0.34 pt	rejected
Test-time mirror averaging	-0.79 pt	rejected
Adaptive OAS covariance shrinkage	-1.05 pt	rejected
Longer optimizer budget	-1.38 pt	rejected
Learnable Sinc filterbank (external)	+0.80 pt	insufficient
Learnable spatial BiMap on top (external)	-0.86 pt	rejected

Table 6 — The full ledger. 'Insufficient' means it improved the model but did not close the external gap to ShallowConvNet.

Two results deserve emphasis. First, **the estimator, not the geometry, was the deficiency**: replacing the SGD head with a convex L2 logistic head on the same frozen-reference tangent features lifts local accuracy from 85.5% to 88.5%, matching a fair non-transductive classical Riemannian pipeline (89.9%). Second, **fixed filter banks lose at scale**: over all 52 external subjects, classical Riemann with the same four fixed bands also loses to ShallowConvNet (59.7 vs 62.9), and neither a learnable Sinc filterbank nor added learnable spatial filtering closed that gap.

9. What these results do and do not support

Supported. A compact (9.7k-parameter, 4.5 ms) geometric decoder with competitive task accuracy under two leakage-controlled protocols; the most accurate deep architecture on this cohort and the most consistent across unseen subjects; a markedly safer commit profile via an independent intent gate and abstention; and a frozen-weight, label-free online adaptation stack with constant memory and per-window auditability.

Not supported. Accuracy superiority over classical FB-CSP or over ShallowConvNet at larger data scale; any benefit from the learned SPD residual or from the online updates on task accuracy (both measured at approximately zero); clinical validity — the cohort is eight healthy volunteers with no disabled or paralyzed participants; and continuous asynchronous or repeated closed-loop performance of this decoder.

Novelty positioning. The components are established: tangent-space classification (Barachant 2012), SPDNet layers (Huang & Van Gool 2017), Euclidean/Riemannian alignment (He & Wu 2020; Zanini 2018), unsupervised bias adaptation (Vidaurre 2011), and learnable Sinc filters (Ravanelli & Bengio 2018). The defensible contribution is not a new classifier but the **deployed system**: the decoupling of covariance recentering from decision-boundary recentering as two bounded label-free causal states, combined with an explicit intent gate and abstention, and evaluated with a safety/availability accounting that the geometric deep-learning literature does not report.

10. Reproduction

Adopted configuration (defaults reproduce the un-augmented baseline exactly):

```
python -m deepnet.experiment benchmark --subjects all \
  --models geoadapt,riemann,fbcspp --seeds 7,17,27 --window deployment \
  --calibration-windows 20 --max-epochs 180 --patience 25 --device cuda \
  --lr-swap-prob 0.5 --select-metric blend
python -m deepnet.dnn_benchmark --archs geoadapt,eegnet,shallow,deep,conformer,atcnet \
  --subjects all --seeds 7,17,27 --augment --device cuda
python -m deepnet.external_benchmark --subjects 1-30 --folds 5 --augment --device cuda
```

All metrics are recomputed from retained per-window traces; aggregates that disagree with their traces cause a hard failure. Result artifacts and the full ledger live under `deepnet/results/`.